



observation



frozen pretrained



analytic, no parameters



trained by the HJ loss



latent state

Multi-view RGB-D

20 Hz

head + 2 wrist cameras
RGB + metric depth
 $K \cdot T(\text{cam} \leftarrow \text{world})$ per view

Frozen vision tower

frozen

Swin-2D + FPN neck (RGB)
Swin-3D + neck (depth)
depth-fusion spatial enhancer
2 of 4 scales (strides 16, 32)

Scene adapter

trained

Linear + RMSNorm
1200 x 256 + world 3D pos

GeometricTrunk (x3)

trained

1 cross-attn: body $\leftarrow$ scene
logit bias MLP($\log|r|$, r_{dir})
+ attn-weighted direction
2 self-attn over 22 tokens
3 FFN 256 $\rightarrow$ 1024 $\rightarrow$ 256
pre-norm RMSNorm residuals
queries: body / keys: scene
 $r = p_{\text{scene}} - p_{\text{body}}$ (metres)

Joint state

q: 14-dim (16 Franka)

Frozen robot-state encoder

frozen

FK $\rightarrow$ [theta, x, y, z, quat]
joint-graph distance attention
 $\rightarrow$ 14 (16) state tokens x 256

RobotInjection

trained

RMSNorm(state) + body cols
shared per-token MLP
+ 2 learned residual tokens
anchored at the grippers
 $\rightarrow$ 22 tokens x 256 + pos

URDF + link poses

velocity limits, L_{reach} ,
collision spheres, chain

Analytic body features

closed form

16 cols/link: p/L_{reach} , quat,
 v , w , chain depth, arm id, grip
 J_{lin} , J_{ang} , joint_index,
 $d\theta_{\text{max}} = \kappa \cdot v_{\text{lim}} \cdot dt$

Encoded latent state

shared by both critics

body 22 x 256 · pos 22 x 3 · mask
scene 1200 x 256 · pos 1200 x 3

J_{lin} , J_{ang} , joint_index, $d\theta_{\text{max}}$ $\rightarrow$ actor and critic heads